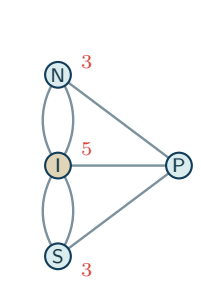
GRAPH THEORY LADDER 4 OF 9

Euler trails and circuits

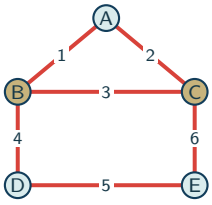
A walk using every edge once exists when at most two vertices have odd degree.

1 / THE RULE AND THE PICTURE

A connected graph has an Euler circuit exactly when every degree is even



Königsberg, degrees $3, 3, 5, 3$



house trail $B \to C$

Euler asked in 1736 whether one could cross each of the seven bridges of Königsberg exactly once, and his answer is

Euler circuit $\iff$ connected, all degrees even

with an Euler trail from one vertex to another when exactly two degrees are odd.

2 / THE THREE CASES

Count the vertices of odd degree

degrees	what happens	example
every degree even	an Euler circuit exists, starting anywhere and returning	K_5, every degree 4
exactly two odd degrees	an Euler trail exists, from one odd vertex to the other	the house, from B to C
more than two odd degrees	no Euler trail at all	Königsberg, four odd vertices; K_4, every degree 3

3 / WHY THE RULE WORKS

Each pass through a vertex uses two edges

Whenever a trail passes through a vertex it arrives along one edge and leaves along another, so the edges at that vertex are used in pairs, and an inner vertex needs an even degree.
Only the first and last vertices of an open trail can have odd degree, and a closed circuit has none.
Königsberg has four odd vertices, with degrees $3, 3, 5, 3$, so no route exists, and one needs to add at least two bridges to pair the odd vertices before a circuit appears.

4 / FINDING A TRAIL

Walk until stuck and splice the loops in

For the house graph start at the odd vertex B and walk B, A, C, B, D, E, C, which uses all six edges and ends at the other odd vertex C, as the numbered edges show.
The general method of Hierholzer starts anywhere, walks without repeating an edge until it is stuck, which in an even graph can only happen at the start, and then splices a further loop into the first unused vertex until every edge is used.
The method is fast, so Euler questions are easy.

5 / TRY THESE

Count the odd vertices first

(a) Does K_5 have an Euler circuit? Every degree is 4, so yes.
(b) Does the cube graph, where every vertex has degree 3, have an Euler trail? No, since it has eight odd vertices.
(c) A postman must walk every street of a graph and return: if it has $2k$ odd vertices he repeats at least k streets, which is the Chinese postman problem.

ASIDE / EDGES AGAINST VERTICES

An Euler trail visits every edge once and a Hamilton cycle visits every vertex once, and the first has a simple test while the second has none.

LINKS TO OTHER SHEETS

The degree rule is on the first sheet, and the next sheet turns to Hamilton cycles on the same house graph.

Convention. A trail repeats no edge; an Euler circuit is a closed trail through every edge; the degrees $3, 3, 5, 3$ are those of the four land masses of Königsberg.